MILITARY SPECIFICATION
SEQUENTIAL TIMER, M7, ASSEMBLY

1. SCOPE

1.1 This specification covers Sequential Timer, M7, Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-E-005272C	-	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-S-45743	-	Soldering, Manual Type, High Reliability, Electrical Connections, for Missile Systems, Procedure For

STANDARDS
Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1167	-	Ammunition Data Card

DRAWINGS
Munitions Command

F8595101	-	Sequential Timer, M7, Assembly
C8811137	-	Container, Metal, Ammunition, for Timer, Sequential: M7
F10520615	-	Carton, (unit pack) Carton, (unit package) Timer, Sequential, XM-7 and XM-8 Assembly, Details, Packing
IEL-8595101	-	Inspection Equipment List for Sequential, Timer, M7, Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

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2.2 Other Publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 General - Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards and specifications, except as provided in 6.1.

3.2 Indicators and drive fitting assembly

3.2.1 Indicators - None of the indicators shall move in excess of 0.025 seconds during any of the prescribed environmental tests, except the tenths indicator (part #8595182) which may not move more than 1.2 seconds in a direction opposite to the setting.

3.2.2 Drive fitting assembly - When the drive fitting assembly is turned forty (40) revolutions in the clockwise direction, the indicators shall not move in excess of 2.0 seconds.

3.3 Activation-movement mounting plate assembly

3.3.1 Arming-safety levers and movements - When the movement mounting plate assembly is set at any time setting between 8.2 and 118 seconds and subjected to an axial acceleration (along Z axis) of 13 g's maximum, the safety levers shall release and the movements shall run (see 6.2.1).

3.3.2 Nonarming-setback releases and safety levers - When the movement mounting plate assembly is set at any time setting between 8.2 and 118 seconds and subjected to an axial acceleration (along Z axis) of 10 g's minimum, the setback release shall not release the safety lever (see 6.2.1).

3.4 Movements and switch assemblies (Timer sub-assembly)

3.4.1 Internal zero setting - The movements and switch assemblies shall be "zero set" as described on Drawing F8595105.

3.5 Electrical properties. - Timer sub-assembly - The timer sub-assembly shall exhibit the electrical properties as required in Table 1.

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3.5.1 Resistance - The base assembly 8595112, assembled with the movement mounting plate assembly 8595149 and the switches and wiring as per schematic F8595768 shall exhibit the electrical properties listed in Table I. Resistance checks up to .11 ohms shall be conducted utilizing a maximum of 20 MA applied current. All 20 megohm checks to utilize 500 ± 5 volts direct current.

TABLE I

Terminals From Pin(s)	Terminals To Pin(s)	Resistance	Switch	Closed on Movement	Timer Setting (secs)
P	Z/X	.11 ohms max	T1	I,II	Not Specified
P	Z	.11 ohms max	T1	I,III	Not Specified
P	Z	.11 ohms max	T1	II,III	Not Specified
P	Z	.11 ohms max	T1	IV	Not Specified
P	Z	.11 ohms max	T1	V	Not Specified
AA/CC	BB/DD	.06 ohms max	T1	I	Not Specified
AA/CC	BB/DD	.06 ohms max	T1	II	Not Specified
AA/CC	BB/DD	.06 ohms max	T1	III	Not Specified
AA/CC	R/T	.11 ohms max	T2	I,II	Not Specified
AA/CC	R/T	.11 ohms max	T2	I,III	Not Specified
AA/CC	R/T	.11 ohms max	T2	II,III	Not Specified
AA/CC	R/T	.06 ohms max	T2	IV	Not Specified
AA/CC	R/T	.06 ohms max	T2	V	Not Specified
R/T	Y/C	.06 ohms max	T2	IV	Not Specified
R/T	Y/C	.06 ohms max	T2	V	Not Specified
R/T	BB/DD	.11 ohms max	T2 T1	I,II I	Not Specified
R/T	BB/DD	.11 ohms max	T2 T1	I,III III	Not Specified
R/T	BB/DD	.11 ohms max	T2 T1	II,III II	Not Specified
R/T	BB/DD	.11 ohms max	T2 T1	IV I	Not Specified
R/T	BB/DD	.11 ohms max	T2 T1	IV II	Not Specified

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3.5.1 (Cont'd)

Terminals From Pin(s)	Terminals To Pin(s)	Resistance	Switch	Closed on Movement	Timer Setting (secs)
R/T	BB/DD	.11 ohms max	T2 T1	IV III	Not specified
R/T	BB/DD	.11 ohms max	T2 T1	V I	Not specified
R/T	BB/DD	.11 ohms max	T2 T1	V II	Not specified
R/T	BB/DD	.11 ohms max	T2 T1	V III	Not specified
J/L	B/D/F	.11 ohms max	Short range T3	Option Switch I,II	Over entire range 8.0 to 120.0
J/L	B/D/F	.11 ohms max	Short range T3	Option Switch I,III	Over entire range 8.0 to 120.0
J/L	B/D/F	.11 ohms max	Short range T3	Option Switch II,III	Over entire range 8.0 to 120.0
J/L	U/W	.11 ohms max	T3	I,II	Not specified
J/L	U/W	.11 ohms max	T3	I,III	Not specified
J/L	U/W	.11 ohms max	T3	II,III	Not specified
J/L	M/K/H	.11 ohms max	Long range T3	Option Switch I,II	Over entire range 26.0 to 120.0
J/L	M/K/H	.11 ohms max	Long range T3	Option Switch I,III	Over entire range 26.0 to 120.0
J/L	M/L/H	.11 ohms max	Long range T3	Option Swtich II,III	Over entire range 26.0 to 120.0
A	R	20 megohms min	All	All	Not specified
A	Z	20 megohms min	All	All	Not specified

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3.5.1 Cont'd

Terminals From Pin(s)	Terminals To Pin(s)	Resistance	Switch	Closed on Movement	Timer Setting (secs)
Z	R	20 megohms min	All	All	Not specified
R	J	20 megohms min	All	All	Not specified
J	Z	20 megohms min	All	All	Not specified
A	MP*	20 megohms min	All	All	Not specified
J	MP*	20 megohms min	All	All	Not specified
R	MP*	20 megohms min	All	All	Not specified
Z	MP*	20 megohms min	All	All	Not specified
B/D/F	J/L	20 megohms min	T3	I,II, III	Over entire range of times greater than 125.0 sec and less than 5.0 sec.
M/K/H	J/L	20 megohms min	T3	I,II III	Over entire range of times greater than 125.0 sec and less than 25.0 sec.

* = Mounting Plate Ground

3.6 Timer assembly

3.6.1 Time setting - Time setting shall be in a counter clockwise direction for countdown of indicator wheels from 000.00 representing 130.00 to 000.01 seconds.

3.6.2 Safety reset - The timer assembly shall be capable of being reset to safe (zero), however, only if no setback release has functioned.

3.6.3 Setting torque - The torque required to set the timer assembly shall not exceed fifteen (15) inch pounds or be less than two (2) inch pounds (applicable at all temperatures between minus 40 degrees and plus 160 degrees Fahrenheit (^{o} F).

3.6.4 Electrical Properties-Timer Assembly

3.6.4.1 Resistance - The timer assembly shall exhibit the electrical properties listed in Table II when the timer is cranked through the entire setting range. Resistance checks up to .15 ohms shall be conducted utilizing a maximum of 20 MA applied current. All 20 megohm checks to utilize 500 $\pm$ 5 volts direct current.

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3.6.4.1 (Cont'd)

TABLE II

Terminals From	Terminals To	Resistance	Timer Setting (seconds)
A/B	C/D	Ohms	
D	B	0.15 ohms max	Any Setting
F	D	.060 ohms max	Any Setting
L	J	0.15 ohms max	Any Setting
N	L	.100 ohms max	Any Setting
N	J	.100 ohms max	Any Setting
K	M	0.15 ohms max	Any Setting
K	H	.060 ohms max	Any Setting
H	M	.060 ohms max	Any Setting
R	T	0.15 ohms max	Any Setting
V	R	.060 ohms max	Any Setting
V	T	.060 ohms max	Any Setting
C	Y	.060 ohms max	Any Setting
S	P	.075 ohms max	Any Setting
W	U	.075 ohms max	Any Setting
X	Z	.060 ohms max	Any Setting
AA	CC	.075 ohms max	Any Setting
BB	DD	.075 ohms max	Any Setting
A	E	.150 ohms max	Check over entire setting range
B/D/F	U/W	0.15 ohms max	Over entire range=8.0 to 120 sec
B/D/F	U/W	20 megohms	Over entire range=125.025 to 130.0 secs "and" 0 to 4.975 secs

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3.6.4.1 (Cont'd)

TABLE II

Terminals From	Terminals To	Resistance	Timer Setting (seconds)
B/D/F	M/K/H	0.15 ohms max	Over entire range=26.025 to 120 secs.
B/D/F	M/K/H	20 Megohms	Over entire range=125.025 to 130.0 secs. and 0 to 24.975 secs.
A	AA	20 megohms min	Any Setting
A	B	20 megohms min	Any Setting
A	BB	20 megohms min	Any Setting
A	J	20 megohms min	Any Setting
A	K	20 megohms min	Any Setting
A	P	20 megohms min	Any Setting
A	R	20 megohms min	Any Setting
A	W	20 megohms min	Any Setting
A	Y	20 megohms min	Any Setting
A	Z	20 megohms min	Any Setting
A	Ground	20 megohms min	Any Setting
AA	B	20 megohms min	Any Setting
AA	BB	20 megohms min	Any Setting
AA	J	20 megohms min	Any Setting
AA	K	20 megohms min	Any Setting
AA	P	20 megohms min	Any Setting
AA	R	20 megohms min	Any Setting
AA	W	20 megohms min	Any Setting

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3.6.4.1 (Cont'd)

TABLE II

Terminals From	Terminals To	Resistance	Timer Setting (seconds)
AA	Y	20 megohms min	Any Setting
AA	Z	20 megohms min	Any Setting
AA	Ground	20 megohms min	Any Setting
B	BB	20 megohms min	Any Setting
B	J	20 megohms min	Any Setting
B	P	20 megohms min	Any Setting
B	R	20 megohms min	Any Setting
B	Y	20 megohms min	Any Setting
B	Z	20 megohms min	Any Setting
B	Ground	20 megohms min	Any Setting
BB	J	20 megohms min	Any Setting
BB	K	20 megohms min	Any Setting
BB	P	20 megohms min	Any Setting
BB	R	20 megohms min	Any Setting
BB	W	20 megohms min	Any Setting
BB	Y	20 megohms min	Any Setting
BB	Z	20 megohms min	Any Setting
BB	Ground	20 megohms min	Any Setting
J	K	20 megohms min	Any Setting
J	P	20 megohms min	Any Setting
J	R	20 megohms min	Any Setting
J	W	20 megohms min	Any Setting
J	Y	20 megohms min	Any Setting

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3.6.4.1 (Cont'd)

TABLE II

Terminals From	Terminals To	Resistance	Timer Setting (seconds)
J	Z	20 megohms min	Any Setting
J	Ground	20 megohms min	Any Setting
K	P	20 megohms min	Any Setting
K	R	20 megohms min	Any Setting
K	Y	20 megohms min	Any Setting
K	Z	20 megohms min	Any Setting
K	Ground	20 megohms min	Any Setting
P	R	20 megohms min	Any Setting
P	W	20 megohms min	Any Setting
P	Y	20 megohms min	Any Setting
P	Z	20 megohms min	Any Setting
P	Ground	20 megohms min	Any Setting
R	W	20 megohms min	Any Setting
R	Y	20 megohms min	Any Setting
R	Z	20 megohms min	Any Setting
R	Ground	20 megohms min	Any Setting
W	Y	20 megohms min	Any Setting
W	Z	20 megohms min	Any Setting
W	Ground	20 megohms min	Any Setting
Y	Z	20 megohms min	Any Setting
Y	Ground	20 megohms min	Any Setting
Z	Ground	20 megohms min	Any Setting

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3.6.5 Timing Accuracy - The timer assembly shall provide the time functions listed in Table III. Each time function shall begin upon breaking of the safety circuit (Pin A to Pin E) and shall end upon completion of the applicable circuit utilizing a maximum of 100 MA applied current. T_s is the set time.

TABLE III

Function	Circuit From Pin(s)	Circuit to Pin(s)	Shall Occur at (seconds)	Applicable Timer Settings
T3A	B and D	J and L	$T_s + 0.2 \pm 0.3$	8.2 up to 25.0
T3B	B and D	J and L	$T_s + .2 \pm (.05 + 0.01\ T_s)$	25.0 up to 118.0 incl.
TO	AA and CC	BB and DD	$(T_s + 0.2\text{-}7)\ ^{+.0.50}_{-1.25}$	8.2 up to 118.0 incl.
T1A	P	Z	$(T_3 - 7)\ ^{+0.40}_{-1.00}$	8.2 up to 25.0 incl.
T1B	P	Z	$(T_s + 2\text{-}14)\ ^{+1.0}_{-1.2}$	25.0 up to 118.0 incl.
T2A	R and T	AA and CC	$(T_3 - 4)\ ^{+1.0}_{-0.5}$	8.2 up to 25.0 incl.
T2B	R and T	AA and CC	$(T_{1B} + 3)\ ^{+1.0}_{-0.3}$	25.0 up to 118.0 incl.

3.6.6 Operated Circuit Resistance - After being subjected to environmental tests and timing accuracy, the timer shall comply with the resistance values listed in Table IV.

TABLE IV

Event	Circuit From Pin	Circuit To Pin	Shall Exhibit	Max Required Value	Under the Following Condition
T_0	AA	BB	Resistance	0.06	Max. applied current 20 MA
T_1	P	Z	Resistance	0.11	Max. applied current 20 MA
T_2	AA	R	Resistance	0.06	Max. applied current 20 MA
T_3	J	B	Resistance	0.11	Max. applied current 20 MA

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3.6.7 Safe setting and backlash removal - The timer assembly to be shipped set on zero (safe setting) and shall have the backlash removed from the setting gear train.

3.6.8 Safety - When the timer assembly is set at 126.0 seconds or set at 004.0 sec and subjected to an axial acceleration (along Z axis) of 50 g's minimum, no setback release shall function (see 6.2.1).

3.6.9 Function - The timer assembly while spinning at 420 revolutions per minute (7 rev per sec) and subjected to an axial acceleration (along Z axis) of 14 g's maximum for twenty (20) milliseconds, minimum at any time setting between 8.2 and 118.0 seconds the timer assembly shall function and demonstrate the required electrical properties and comply with the accuracy requirements of Table III as applicable within the temperature range of minus 40 degrees to plus 160 degrees Fahrenheit (see 6.2.1).

3.6.9.1 Long range - Timing elements No. 1,2,3,4 and 5 shall function when the timer assembly is set at any time setting between twenty-three (023.0) and one hundred and eighteen (118.0) seconds (see 6.3).

3.6.9.2 Short range - Timing elements No. 4 and 5 shall not function when the timer assembly is set for less than eighteen (018.0) seconds (see 6.3).

3.6.10 Nonfunction - The timer assembly while spinning at 420 RPM and subjected to an axial acceleration (along Z axis) of 10 g's minimum, at any time setting between 8.2 and 118.0 seconds no timing element shall function within the temperature range of minus 40 degrees to plus 160 degrees Fahrenheit (see 6.2.1 and 6.3).

3.7 Environmental - Timer assembly set on zero, ambient temperature

3.7.1 Shock - The timer assembly when subjected to shock in accordance with MIL-E-005272C, paragraph 4.15.5, except that the acceleration shall be 20 g's plus or minus 2 g, shall demonstrate the required electrical properties and shall comply with the requirements in 3.2.1, 3.6.3, 3.6.9, 3.6.10 and Table III (see 4.2.5).

3.7.2 Vibration - The timer assembly when subjected to vibration in accordance with Table V shall demonstrate the required electrical properties and shall comply with the requirements as specified in 3.2.1, 3.6.3, 3.6.9, 3.6.10 and Table III (see 4.2.5 and 4.3.3.12).

TABLE V

Frequency Cycles per Second	Double Amplitude or peak g's + 10%	Cycle Time in min.	Test Time	Axis
10-20-10	2g's	3 + .5	30 minutes	X
10-20-10	2g's	3 + .5	30 minutes	Y

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3.7.2 (Cont'd)

TABLE V

Frequency, Cycles per Second	Double Amplitude or peak g's + 10%	Cycle Time in min.	Test Time	Axis
20-50-20	0.036 inches	3 + .5	30 minutes	X
20-50-20	0.036 inches	3 + .5	30 minutes	Y
50-100-50	5 g's	3 + .5	15 minutes	X
50-100-50	5 g's	3 + .5	15 minutes	Y
100-500-100	5 g's	3 + .5	10 minutes	X
100-500-100	5 g's	3 + .5	10 minutes	Y
75-1000-75	5 g's	3 + .5	20 minutes	Z

3.8 Dry nitrogen purge - The timer assembly shall be purged with dry nitrogen (dew point -75 F or lower) prior to the pressure retention test (see Dwg 8595101).

3.9 Pressure retention - The timer assembly shall not leak more then 0.25 pounds per square inch pressure in 15 minutes when subjected to a pressure of 12 to 15 pounds per square inch gage at ambient conditions (see 6.2.1.1).

3.10 Workmanship - The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects - All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.10.1.1 Firing arm assembly - The firing arm assembly shall contain no fracture at the bend.

3.10.2 Plastic defects

3.10.2.1 Wafers, plates, spacers and rotors - All components shall be free from fractures, burrs, flaking of gold plate on clips and wipers.

3.10.2.2 Cables - All cables shall be free from blisters, wrinkles and splits.

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3.10.2.3 Solvents - No solvent containing aromatic hydrocarbons, chlorinated hydrocarbons, and alkaline substances shall be used in conjunction with any plastic parts.

3.10.3 Soldering - The soldering shall be uniform and electrically conductive. The joints shall have a smooth metallic appearance and shall show no evidence of excessive solder, mounds, presence of flux, foreign matter or having been overheated (slight discoloration is permissible) and insulation adjacent to the soldering area shall not be damaged. The soldering procedure and certification of soldering operators shall be in accordance with MIL-S-45743. A copy of the certification shall be furnished by the processing activity.

3.10.4 Threads - Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.5 Burr - No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.6 Surface finish - The requirements for surface finish are as detailed on the applicable drawing.

3.10.7 Protective finish - Protective finish shall be in accordance with the applicable specifications and drawings, and shall apply to all surfaces of the component regardless of configuration.

3.10.8 Foreign matter - No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The methods and agents used in processing and cleaning shall not adversely affect the finish or contaminate the parts.

3.11 Serialization - A serial number shall be assigned and affixed by the contractor to each assembly or part when required by the applicable drawing. The serial number shall be in addition to the lot number, but in no event shall there be any possibility of duplication nor any indication of order or size of production quantity in either lot designation, serial number or any other number.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions - Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Lot formation - Each inspection lot of fuzes shall be comprised of components (except explosive components) each produced by one manufacturer, by one process, in accordance with the same drawing and revision and specification and revisions, as applicable, and assembled on one assembly line. Each explosive component within a lot of assembled fuzes shall be limited to one manufacturer and one interfix number.

4.2 First article sample - When required, the first article sample shall be as specified in the contract and shall consist of the following:

260 (aggregate) component parts as delineated in 4.2.2

700 (aggregate) subassemblies as delineated in 4.2.3

20 (aggregate) timer assemblies as delineated in 4.2.5

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and Test - After inspection by the contractor and provisional acceptance at source by the Government, two hundred and sixty (260) component parts, 700 subassemblies and 20 timer assemblies shall be inspected, as provided herein, at a government laboratory or such other facility as specified in the contract.

4.2.2 Components

4.2.2.1 Twenty (20) of each of the (12) components shall be tested as in 4.3.3.2 (Hardness) and 4.3.3.3 (load).

4.2.2.2 Twenty (20) of each component (1) shall be tested as in 4.3.3.4 (compression).

4.2.3 Subassemblies

4.2.3.1 Twenty-five (25) of each of the (14) subassemblies shall be tested as in 4.3.3.5.1 (Static load).

4.2.3.2 Twenty-five (25) of each of the (14) subassemblies shall be tested as in 4.3.3.5.2 (Torque).

4.2.4 Acceptance Criteria - Failure of any component or subassembly to comply with the requirements of the applicable drawing shall be cause for rejection of the first article sample.

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4.2.5 Timer assemblies (non-destructive-see 6.4) - Twenty (20) timers (100 timing elements-see 6.3) shall be subjected to the following environmental phases in the sequence shown.

	Phase & Sequence	Quantity	Time setting-seconds	Method
1	1. Non-function (ambient)	All	50 seconds	4.4.18
	2. Reset to Zero	All	Zero	----
	3. Check for closed circuit	All	----	Pins A to E
	4. Shock (ambient)	8	Zero	4.4.20.1
	5. Vibrations (ambient)	*	Zero	4.4.20.2
	6. Function & Accuracy (amb)	*	4 at 10sec & 4 at 100sec	4.4.19
	Operated circuit resistance	*	3 at 10sec & 3 at 100sec	4.4.15
1	7. Function & Accuracy (-40° F)	6	3 at 10sec & 3 at 100sec	4.4.19 & 4.4.20.3
	Operated circuit resistance	-		4.4.15
1	8. Function & Accuracy (+160° F)	6	3 at 10sec & 3 at 100sec	4.4.19 &
	Operated circuit resistance	-		4.4.20.3 & 4.4.15

* (Same units used for shock)

1 Note: (Check timer setting torque on all units when setting for time).

4.2.5.1 Acceptance criteria - The occurance of any of the following shall be cause for rejection of the first article sample:

1. One or more timers fail to comply with any one of the following requirements:

   a. Nonfunction - 3.6.10
   b. Accuracy - Table III
   c. Operated circuit resistance - Table IV
   d. Timer setting torque - 3.6.3
   e. Indicator setting - 3.2.1

2. More than one timing element in any one timer fails to function (see 6.3).

3. Failure of more than four (4) timing elements to function in the first article sample.

4.3 Inspection provisions

4.3.1 Submission of product - The product shall be presented in accordance with MIL-STD-105.

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4.3.2 Examination - One hundred percent examination shall be performed for all critical defects. Examination for major defects shall be performed on a single defect basis in accordance with MIL-STD-105. For each classification of defects contained herein, the AQL given shall be applied to each major defect listed. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.3.2.1 Base Drawing F8595174 covering a detail of Drawing F8595101

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR	AQL = 0.40%	
101.	Diameter of M hole incorrect	Gage
102.	Diameter of No. 9 hole, incorrect	Gage
103.	Diameter of No. 5 hole, incorrect	Gage
104.	Location of No. 5 hole, incorrect	Gage
105.	Location of mounting holes incorrect	Gage
106.	Overall height exceeds max.	Gage
107.	Holes No. 6,7 and 8 broken thru	Visual
108.	Flatness of internal surface with datum "B" incorrect	Gage
109.	Surface finish, improper	Visual
110.	Threads damaged or not full	Visual

4.3.2.2 Plate, Movement Mounting Drawing F8594290 covering a detail of Drawing F8595101

CRITICAL		
	None defined	
MAJOR	AQL = 0.40%	
101.	Diameter of No. 74 hole exceeds max.	Gage
102.	Diameter of center bosses incorrect	Gage
103.	Location of hole No. 92, 93, 94, 95 or 96 incorrect	Gage
104.	Thickness of plate incorrect	Gage
105.	Surface finish improper (large cut out area, boss shoulder and hole)	Visual

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4.3.2.3 Wiring Schematic Drawing F8595768 covering a detail of Drawing F8595101

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 0.40%	
101. Wire covering cut or frayed	Visual
102. Soldering inadequate, missing, excessive or charred (wire or cover)	Visual
103. Wiring insecure	Manual/Visual
104. Foreign matter	Visual

4.3.2.4 Movement Mounting Plate Assembly Drawing F8595149 covering a detail of Drawing F8595101

CRITICAL	
1. Escapement lever pallet(s) missing, damaged or deformed	Visual
2. Release ring spring(s) missing or broken	Visual
3. Safety lever spring(s) missing, loose or broken	Manual/Visual
MAJOR AQL = 0.40%	
101. Distance from bottom of release ring to bottom of safety lever incorrect	Gage
102. Release ring contact screw improperly adjusted	Gage
103. Release ring improperly aligned	Gage
104. Release ring adjusting screw not adjusted properly for minimum play when setting gear assembly is at "000" position	Gage
105. Arbor(s) not prewound	Visual
106. Movement(s) not fully wound	Manual/Visual
107. Burr	Visual
108. Foreign matter	Visual

4.3.2.5 Setting Shaft Mounting Plate Assembly Drawing F8595736 covering a detail of Drawing F8595101

CRITICAL

None defined

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 0.40%	
101. Excessive end shake in indicator(s)	Gage
102. Dowel pin extension exceeds max.	Gage
103. Improper assembly	Visual
104. Numeral(s) not clear	Visual
105. Gears damaged	Visual
106. Foreign matter	Visual
107. Sealant missing or excessive	Visual

4.3.2.6 Base Assembly Drawing F8595112 covering a detail of Drawing F8595101

CRITICAL	
None defined	
MAJOR AQL = 0.40%	
101. Wire covering cut or frayed	Visual
102. Soldering inadequate, missing, excessive or charred (wire, clip or cover)	Visual
103. Improper assembly	Visual
104. Burr	Visual
105. Foreign matter	Visual

4.3.2.7 Timer Subassembly Drawing F8595105 covering a detail of Drawing F8595101

CRITICAL	
1. Firing arm fractured	Visual
2. Firing arm spring missing, broken or hooked up improperly	Visual
MAJOR AQL = 0.40%	
101. Dimension under center gear bearing retainer incorrect	Gage
102. Distance from top of timing disc to bottom of firing arm(s) incorrect	Gage
103. Distance between safety and timing disc incorrect	Gage
104. Play in setback release ring	Manual/Visual
105. Bearing edge of firing arm #1 not aligned with safety line on movements I, II and III	Visual
106. Bearing edge of firing arm #2 not aligned with safety line on movements IV and V	Visual
107. Improper assembly (timing disc on wrong movement)	Visual

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4.3.2.7 (Cont'd)

	Defect	Method of Inspection
108.	Zero setting improper	Visual
109.	Numbers on dial(s) missing or not clear	Visual
110.	Burr	Visual
111.	Foreign matter	Visual

4.3.2.8 Sequential Timer, M7 Assembly Drawing F8595101

CRITICAL

1.	Bent or broken connector pin(s)	Visual

MAJOR AQL = 0.40%

101.	Overall height exceeds max.	Gage
102.	Timer not set on zero	Visual
103.	Window damaged or not clear	Visual
104.	Valve, plastic seal broken (to be checked after pressure test)	Visual
105.	Marking missing, incorrect or illegible	Visual
106.	Identification plate(s) loose or missing	Visual
107.	Sealing missing or excessive (to be checked after pressure test)	Visual
108.	Foreign matter	Visual
109.	Exterior surface(s) damaged	Visual

4.3.2.9 Carton (inner pack) Drawing F10520615

CRITICAL

None defined

MAJOR AQL = 0.65%

101.	Filler missing	Visual
102.	Improper assembly	Visual
103.	Contents loose	Manual/Visual
104.	Foreign matter	Visual
105.	Carton improperly closed or sealed	Visual

4.3.2.10 Carton (outer package) Drawing F10520615

CRITICAL

None defined

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4.3.2.10 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 0.65%	
101. Filler missing	Visual
102. Marking misleading or unidentifiable	Visual
103. Contents loose	Manual/Visual
104. Carton improperly closed or sealed	Visual

4.3.2.11 Metal Container Drawing C8811137

CRITICAL

None defined

MAJOR

101. Wrapper inadequate or missing	Visual
102. Bag damaged or not sealed	Visual
103. Contents loose	Visual
104. Carton not sealed adequately	Visual
105. Carton markings missing, incorrect or illegible	Visual
106. Evidence of moisture in padding	Manual/Visual
107. Pad(s) damaged or incorrect	Visual
108. Container damaged	Visual
109. Container markings missing, incorrect or illegible	Visual
110. Metallic seal missing or loose	Visual

4.3.2.12 Process control audit - Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings.

A. Drawing F8595768 and F8595112

1. Soldering techniques
2. Cleaning solvents used
3. Application of sealant and drying

B. Drawing B10522365

1. Pellet pull test

C. Drawing F8595149

1. Lubrication and application of grease
2. Arbor prewound
3. Regulation
4. Setting gear torque
5. Application of sealant and drying

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4.3.2.11 (Cont'd)

D. Drawings F8595736 and F8596397

1. Lubrication

E. Drawing F8595105

1. Timing disc torque
2. Safety (zero) torque (after assembly of locknut)
3. Zero setting

F. Drawing F8595101

1. Timer setting torque
2. Torque valve cap (after purging)
3. Torque cover screws
4. Application of sealant and drying
5. Pressure retention
6. Backlash removal (just prior to packaging)

4.3.3 Inspection testing - The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish - The number of specimens or samples (as applicable) and acceptance criteria for protective finish tests shall be in accordance with the applicable standards or specification listed on the drawing (see 4.4.1).

4.3.3.2 Hardness (destructive) - Twenty (20) samples from each lot of the following components shall be subjected to the hardness test. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot. Where plating is required, the sample parts submitted shall be unplated (see 4.4.2).

Components	Drawing Number
Gear, Escapement	C10542765
Lever, Escapement	D10522607
Safety, Zero	D8594343
Safety, (23 Sec.)	D8595726
Spring, Timing Disc (see 4.3.4.4.1)	B8595743
Spring, Tensioning, Setting Gear	C8594347
Shaft	C8595701
Shafts	B8595705 & B8595716
Coupling, Shaft	C10521038
Guide, Adjusting	B8594273
Worm	C8595193

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4.3.3.3 Load - Timing Disc Spring (destructive) - The same 20 timing disc springs subjected to the hardness test shall be subjected to the load specified on Drawing B8595743. Failure of any spring to comply with the requirement shall be cause for rejection of the lot of timing disc springs (see 4.4.3).

4.3.3.4 Compression - Release ring spring (destructive) - Twenty (20) release ring springs from each lot shall be subjected to the compression specified on Drawing C8594293. Failure of any spring to comply with the requirement shall be cause for rejection of the lot of release ring springs (see 4.4.4).

4.3.3.5 Static load and torque

4.3.3.5.1 Static load - Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the static load test. Failure of any subassembly to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (see 4.4.5 and 6.5).

Static Load	Drawings
Gear Assembly	C8595672
Safety Lever Assembly	D8597128
Escapement Gear and Pinion Assembly	C10522613
Escapement Arbor Assembly	F10522611
Mainspring Arbor Assembly	D8595150
No. 1 Gear and Pinion Assembly	C10522608
No. 2 Gear and Pinion Assembly	C10522609
No. 3 Gear and Pinion Assembly	C10522610
No. 4 Gear and Pinion Assembly	C10522612
Worm Gear and Shaft (nondestructive)	C8595804
Timing Disc Subassembly (rivet)	C8594331 & C8595718
Setting Gear Assembly (nondestructive)	D8594296
Safety Release Ring Braket (nondestructive)	C8596141

4.3.3.5.2 Torque - Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the torque test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (see 4.4.5 and 6.5).

Torque	Drawing
Gear Assembly	C8595672 & C8595685
Safety Lever Assembly	D8597128
Escapement Gear and Pinion Assembly	C10522613
Escapement Lever Assembly	C10522611
Safety Assembly	B8595724 & B8595725
Mainspring Arbor Assembly	D8595150
Setting Gear Assembly (nondestructive)	C8594296
Worm and Shaft Assembly (nondestructive)	C8595804
Timing Disc Subassembly (rivet)	C8594331 & C8595718
Timing Disc Assembly (nondestructive)	C8594330 & C8595719

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4.3.3.6 Indicators and drive fitting assembly

4.3.3.6.1 Indicators (see 4.2.5 and 4.3.3.16-nondestructive) - Each timer assembly subjected to a test shall be examined to determine if the indicators have moved from the setting during the course of testing. Failure to comply with the requirement of paragraph 3.2.1 shall be cause for rejection of the first article sample or production lot, as applicable (see 4.4.6.1).

4.3.3.6.2 Drive fitting assembly (nondestructive) - Each drive fitting in each timer assembly produced shall be checked to determine that the indicators are settable in one direction only. Failure to comply with the requirement of paragraph 3.2.2 shall be cause for rejection of the assembly (see 4.4.6.2).

4.3.3.7 Activation-movement mounting plate assembly

4.3.3.7.1 Arming-safety lever and movement (nondestructive) - Each movement mounting plate assembly produced shall be tested to determine compliance with the arming requirement of paragraph 3.3.1. Failure of any movement to run after release of the safety lever shall be cause for rejection of the movement mounting plate assembly (see 4.4.7.1).

4.3.3.7.2 Nonarming-setback release and safety lever (nondestructive) - Each movement mounting plate assembly produced shall be tested to determine compliance with the nonarming requirement of paragraph 3.3.2. Failure of any setback release to retain the safety lever shall be cause for rejection of the movement mounting plate assembly (see 4.4.7.2).

4.3.3.8 Regulation-movement mounting plate assembly (nondestructive) - After adjusting and rewinding each movement assembly produced shall be subjected to two consecutive runs for 120 seconds each. The tapes of each movement shall be inspected to determine compliance with the requirement of Drawing F8595149. Failure to comply with the requirement shall be cause for rejection of the movement mounting plate assembly (see 4.4.8).

4.3.3.9 Movement and switch assemblies (Timer-subassembly)

4.3.3.9.1 Internal zero setting (nondestructive) - Each movement and each switch assembly produced shall be checked for proper zero set. Failure of any unit to comply with the requirement of 3.4.1 shall be cause for rejection of the timer sub-assembly (see 4.4.9.1).

4.3.3.10 Torque-Safety assemblies (nondestructive) - Each safety assembly on each movement assembly produced shall be subjected to the torque specified on Drawing F8595105. Failure of any safety assembly to comply with the requirement shall be cause for rejection of the timer sub-assembly (see 4.4.10).

4.3.3.11 Electrical properties - Timer sub-assembly

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4.3.3.11.1 Resistance (nondestructive) - Each base assembly with movement mounting plate assembly, switches and wiring produced shall be tested to determine compliance with the requirements of paragraph 3.5.1 and Table I. Failure to comply with the requirements shall be cause for rejection of the timer subassembly (see 4.4.11.1).

4.3.3.12 Timer assembly

4.3.3.12.1 Time setting (nondestructive) - Each timer assembly produced shall be checked to determine compliance with the requirement of paragraph 3.6.1 Failure to comply shall be cause for rejection of the timer assembly (see 4.4.12.1).

4.3.3.12.2 Safety reset (nondestructive) - Each timer assembly produced shall be tested to determine compliance with the requirement of paragraph 3.6.2. Failure to comply shall be cause for rejection of the timer assembly (see 4.4.12.2).

4.3.3.12.3 Setting torque (nondestructive) - Each timer assembly produced shall be tested to determine compliance with the requirement of paragraph 3.6.3. Failure to comply with the requirement shall be cause for rejection of the timer assembly (see 4.4.12.3).

4.3.3.13 Electrical properties - Timer assembly

4.3.3.13.1 Resistance (nondestructive) - Each timer assembly produced shall be tested to determine compliance with the requirements of paragraph 3.6.4.1 and Table II. Failure to comply with the requirements shall be cause for rejection of the timer assembly (see 4.4.13.1).

4.3.3.14 Timing accuracy (nondestructive) - The twenty (20) timer assemblies, from the first article sample (4.2.5) and eight (8) timer-assemblies from each production lot (4.3.3.23) shall be tested to determine compliance with the requirement of Table III. Failure to comply with the requirement shall be cause for rejection of the first article or production lot, as applicable (see 4.4.14).

4.3.3.15 Operated circuit resistance (nondestructive) - The timer assemblies which have been subjected to environmental phases and functioned for accuracy shall be tested to determine compliance with the requirement of Table IV. Failure to comply with the requirement shall be cause for rejection of the first article or production lot, as applicable (see 4.2.5, 4.3.3.23 and 4.4.15).

4.3.3.16 Safe setting and backlash removal (nondestructive) - Each timer assembly produced shall be checked to determine compliance with the requirements of paragraph 3.6.7. Failure to comply shall be cause for rejection of the timer assembly (see 4.4.16).

4.3.3.17 Safety (nondestructive) - Each timer assembly produced shall be tested to determine compliance with the requirements of paragraph 3.6.8. Failure to comply with the requirements shall be cause for rejection of the timer assembly (see 4.4.17).

4.3.3.18 Function-Long and Short Range - (see 4.4.19)

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4.3.3.18.1 First article - (see 4.2.5 and 4.2.5.1)

4.3.3.18.2 Production lots - (see 4.3.3.23 and 4.3.3.23.1)

4.3.3.19 Nonfunction (nondestructive) - Each timer assembly produced shall be tested to determine compliance with the requirements of paragraph 3.6.10. Failure to comply with the requirements shall be cause for rejection of the timer assembly (see 4.4.18).

4.3.3.20 Environmental (nondestructive)

4.3.3.20.1 Shock (see 3.7.1, 4.2.5 and 4.4.20.1)

4.3.3.20.2 Vibration (see 3.7.2, 4.2.5 and 4.3.3.23, 4.4.20.2)

4.3.3.20.3 Extreme temperature tests (see 4.2.5, 4.3.3.23 and 4.4.20.3)

4.3.3.21 Dry nitrogen purge (nondestructive) - Each timer assembly produced shall be subjected to a dry nitrogen purge in accordance with Drawing 8595101 and paragraph 3.8. Failure to comply with the requirements shall be cause for rejection of the timer assembly (see 4.4.21).

4.3.3.22 Pressure retention (nondestructive) - Each timer assembly produced shall be subjected to the pressure test in accordance with paragraph 3.9. Failure to comply with the requirement shall be cause for rejection of the timer assembly (see 4.4.22).

4.3. 3.23 Production lots (nondestructive - see 6.4)

4.3.3.23.1 Timer assemblies - Eight (8) timers (40 Timing elements see 6.3) from each production lot shall be subjected to the environmental phases in the sequence shown:

Phase & Sequence	Quantity	Time Setting - Seconds	Method
1. Nonfunction (ambient)	All	50 Seconds	4.4.18
2. Reset to Zero	All	Zero	-
3. Check for Closed Circuit	All	-	Pins A & E
4. Vibration (ambient)	8	Zero	4.4.20.2
1 5. Function & Accuracy (-40° F)	*	4 at 50	See 4.4.19 & 4.4.20.3
Operated Circuit Resistance	*	-	4.4.15
1 6. Function & Accuracy (+160° F)	*	4 at 100 Sec	4.4.19 & 4.4.20.3
Operated Circuit Resistance	*	-	4.4.15

*(Same units used for Vibration)

1 Note: (Check timer setting torque on all units when setting for time)

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4.3.3.23.1.1 Acceptance criteria - The occurance of any of the following shall be cause for rejection of the production lot.

1. One or more timers fail to comply with any one of the following requirements:

a. Nonfunction - 3.6.10
b. Accuracy - Table III
c. Operated Circuit Resistance - Table IV
d. Timer Setting Torque - 3.6.3
e. Indicator Setting - 3.2.1

2. More than one timing element in any one timer fails to function (see 6.3).

3. Failure of more than two (2) timing elements to function in the entire sample.

4.3.4 Inspection equipment - The examination and tests shall be made using equipment listed in IEL-8595101. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures

4.4.1 Protective finish - Test procedures for protective finish shall be in accordance with the applicable specification or MIL-STD referenced on the component drawing.

4.4.2 Hardness - Sample shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three (3) hardness readings taken on each sample unit shall be used as the individual value.

4.4.3 Load (timing disc spring) - The test shall be performed using the values as specified on Drawing B8595743. The timing disc springs that have been subjected to the load test shall be scrapped.

4.4.4 Compression - The release ring springs shall be compressed as specified on Drawing C8594293. The springs that have been subject to the compression test shall be scrapped.

4.4.5 Static load and torque - Escapement lever assemblies shall be tested prior to insertion of the escapement lever spring. The test shall be performed using the values as specified on the applicable drawing. All samples tested shall be scrapped, except the setting gear assembly, worm and shaft assembly, timing disc assembly and safety release ring bracket (see 6.5).

4.4.6 Indicators and drive fitting assembly

4.4.6.1 Indicators - After completion of the shock and vibration test phases, the indicators shall be examined to determine compliance with the requirement of 3.2.1.

4.4.6.2 Drive fitting assembly - The test shall be performed on the timer assembly prior to removal of the backlash in the gear train.

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4.4.7 Activation-movement mounting plate assembly

4.4.7.1 Arming-safety lever and movements - The test shall be performed as follows:

a. Set movement mounting plate assembly between 8.2 and 118.0 seconds, utilizing a simulated setting mechanism, (indicators, etc).

b. Mount the plate on a centrifuge (see 6.2.1).

c. Axial acceleration along Z axis of 13 g's maximum, for two (2) seconds, maximum.

d. Examine to determine compliance with the requirement of paragraph 3.3.1.

e. Rewind the movements and reset the setback releases prior to the nonarming test.

4.4.7.2 Nonarming-setback release and safety lever - The test shall be performed as follows:

a. Nonarming test to be performed after the arming test.

b. Set movement plate assembly between 8.2 and 118.0 seconds, utilizing a simulated setting mechanism.

c. Mount the plate on a centrifuge (6.2.1).

d. Axial acceleration along Z axis of 10 g's minimum for five (5) seconds, minimum.

e. Examine to determine compliance with the requirement of paragraph 3.3.2.

4.4.8 Regulation - The movements shall be regulated using the operational procedure furnished with the regulation equipment. Time keeping and performance shall be checked over a period of not less than 120 seconds. No adjustment shall be made on the escapement during the two consecutive 120 second check runs. After final rewind the movement will be locked and further running of the movement is not permitted.

4.4.9 Movement and switch assemblies (Timer sub-assembly)

4.4.9.1 Internal zero setting - The movements and switches shall be zero set as specified on Drawing 8595105.

4.4.10 Torque-Safety assemblies - The test shall be performed on the timer sub-assembly. The load shall be applied to the safety assemblies 8595724 and 8595725 in a clock wise direction using the values and method as specified on Drawing 8595105.

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4.4.11 Electrical properties - Timer sub-assembly

4.4.11.1 Resistance - The timer sub-assembly shall be subjected to the conditions and tests listed in Table I and exhibit the electrical properties therein under ambient conditions. Prior to performing the measurements of resistance, all timing and arming switches shall have been in the unoperated position for 24 hours minimum. For any combination of closed switches specified in Table I, the circuit resistance shall be measured and recorded for the first closure of each of the switches specified. Movement of a switch shall be restricted to those closures required by Table I, one closure and one opening per combination (Read and record data).

NOTES: 1. A/B to C/D is accomplished by checking (1) A to C or (2) A to D or (3) A to C and D* or (4) B to C or (5) B to D or (6) B to C and D* or by shorting in test equipment A and B and checking to (7) C or (8) D or (9) C and D*.

*Where C and D are shorted in test equipment.

2. D. C. potential applied for all tests required by this table. Positive side of the applied voltage shall be impressed on the terminal(s) listed in the "From Pin(s)" column.

3. High resistance (20 megohms minimum) between terminals and ground may be determined by shorting all terminals and checking directly to ground connection. (Ground is the movement mounting plate. Protective finish must be penetrated at contact point).

4.4.12 Timer assembly

4.4.12.1 Time setting - The test shall be performed during the checks for resistance in accordance with Table II.

4.4.12.2 Safety reset - The test shall be performed at the completion of the checks for resistance in Table II.

4.4.12.3 Setting Torque - The test shall be performed just prior to final zero set and backlash removal from the gear train. All timers from the first article and all timers from each production lot sample shall be checked, when set for time, to determine compliance with the requirement. Record data.

4.4.13 Electrical properties - Timer assembly - Table II

4.4.13.1 Resistance - The test shall be performed as specified in Table II and all readings to be recorded.

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4.4.13.1 (Cont'd)

NOTES: 1. A/B to C/D is accomplished by checking (1) A to C or (2) A to D or (3) A to C and D* or (4) B to C or (5) B to D or (6) B to C and D* or by shorting in test equipment A and B and checking to (7) C or (8) D or (9) C and D*.

*Where C and D are shorted in test equipment.

2. D. C. potential applied for all tests required by this table. Positive side of the applied voltage shall be impressed on the terminal(s) listed in the "From Pins" column.

3. High resistance between terminals and ground may be determined by shorting all terminals and checking directly to ground connection. (Ground is the base or movement mounting plate. Protective finish must be penetrated at contact point).

4.4.14 Timing accuracy (see 4.2.5 and 4.3.3.23) - The time durations which elapse between breaking of the release ring circuit and completion of the circuits associated with the events specified in Table III shall be measured for conformances to the requirements. Accuracy of timers to be read to nearest tenth of a second and all data to be recorded.

4.4.15 Operated circuit resistance - After completion of function and accuracy tests, the timer assemblies shall be checked to determine compliance with the requirements of Table IV. Record data (see 4.2.5 and 4.3.3.23).

4.4.16 Safety setting and backlash removal - The timer assembly shall have the backlash of the gear train removed and set on zero (safe) just prior to packaging.

4.4.17 Safety - The test shall be performed as follows:

a. Mount timer assembly on a centrifuge (see 6.2.1).

b. Set at 126.0 seconds and subject to 50 g's for a duration of 5 seconds, minimum.

c. Check pins A & E for breaking of the release ring circuit.

d. Reset at 004.0 seconds and subject to 50 g's for a duration of 5 seconds.

e. Check pins A & E.

f. Reset to zero.

4.4.18 Nonfunction - The test shall be performed as follows:

a. Mount timer assembly on a centrifuge (see 6.2.1).

b. Set at 50 seconds and subject to 10 g's minimum for a duration of 5 seconds, minimum.

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4.4.18 (Cont'd)

c. Check pins A & E for breaking of the release ring circuit.

d. Reset to zero.

4.4.19 Function-long and short range - The test shall be performed using the operational procedures furnished with the equipment.

a. The function test shall be performed after the nonfunction test.

b. First article (see 4.2.5) for phases and setting time.

c. Production lot samples (see 4.3.3.23) for phases and setting time.

d. Use of Hyge linear accelerator with a spin fixture capable of 420 RPM (see 6.2.1).

e. Power supply with electrical hookup for accuracy checks in Table III. Read to nearest tenth of a second and all data to be recorded.

f. Subject to 14 g's maximum, for 20 milliseconds, minimum.

g. Check operated circuit resistance, Table IV.

h. Post mortem to be performed on each timer to determine element functions (see 6.3).

4.4.20 Environmental

4.4.20.1 Shock - The timer assemblies from the first article (see 4.2.5) shall be tested in accordance with the procedure specified in Specification MIL-E-005272C, paragraph 4.15.5, except that the acceleration shall be 20 g plus or minus 2 g. Upon completion of the shock schedule, examine the indicators for movement (see 3.2.1), but no disassembly to be performed on the timer assembly. The same units tested for shock shall be subjected to a vibration.

4.4.20.2 Vibration - The timer assembly shall be subjected to a series of vibration according to the schedule in Table V. After which, the timer shall be temperature conditioned (see 4.2.5 and 4.3.3.23) and then functioned per paragraph 4.4.19.

4.4.20.3 Extreme temperature tests - The sample timer assemblies shall be functioned as specified in paragraph 3.6.9 under the conditions delineated in paragraph 4.2.5. and 4.3.3.23.

4.4.20.3.1 Temperature conditioning (-40° F and + 160° F) - The test chamber shall be at ambient temperature at the start of the test and the chamber maintained at a temperature of minus 40° F or plus 160° F as applicable, until stabilization of the timer temperature is reached (see 6.6). After stabilization, the timer shall be saturated at minus 40° F or plus 160° F as applicable, for one hour, minimum.

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4.4.21 Dry nitrogen purge - The procedure shall be in accordance with Drawing 8595101 and paragraph 3.8.

4.4.22 Pressure Retention - Prior to the specified pressure retention test, the plug in the pressure retention access hole shall be removed and dry nitrogen (dew point -75 F or lower) of between 2 and 4 psig inserted thru valve in top of cover and maintained while removing the plug. Immediately following the removal of the plug, the pressure shall be increased to 9 psig and maintained for one minute flush. Upon completion of the flush, the pressure shall be reduced to between 2 and 4 psig while reassembling the plug. The complete timer assembly shall then be subjected to a pressure differential not to exceed 15 psi nor below 12 psi. The greater pressure shall be exerted inside the timer. The change in pressure inside the timer that occurs while the timer is subjected to the specified pressure differential for 15 minutes shall be determined. The 15 minutes test interval shall start immediately after the specified pressure differential is obtained. Any change in the pressure inside the timer indicates a leak which must be measured to determine conformance to the requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Preservation and packing shall be level "A" or level "C", as specified (see 6.8).

5.1.1 Level "A" - Packaging, packing and marking shall be in accordance with Drawing C8811137.

5.1.2 Level "B" - Not applicable.

5.1.3 Level "C" - Packaging, packing and marking shall be in accordance with Drawing F10520615.

6. NOTES

6.1 Dimensions

6.1.1 Compliance with tests (proving ground, ballistic performance, and shop) required by the specifications and drawings is mandatory. The contractor will not assume, nor does the government guarantee, that all possible combinations permitted by the tolerance limits of the specifications and drawings will consistently satisfy the test requirements. Therefore, the manufacturer is obligated to choose those combinations of tolerances and fits within the limits of the specifications and drawings that best suit his process needs and still satisfy the testing requirements.

6.1.2 Dimensions which are specifically referred to in the classification of defects, as listed in Section 4, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate fuze performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on fuze functionability.

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6.1.2 Cont'd

This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, and does not constitute authority to establish new dimensions or to relax efforts to achieve the required sizes within the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

6.1.3 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned cutody of the technical data.

6.2 Definitions

6.2.1 Axis

Y axis is parallel to the setting shaft, Drawing C8595122.
X axis is perpendicular to the YZ plane.
Z axis is perpendicular to the plane of the base, Drawing F8595174.

6.2.1.1 Ambient - 75 degrees plus or minus 15 degrees Fahrenheit.
1 plus or minus 1/4 atmosphere pressure.
60 percent max. relative humidity.

6.2.1.2 Arming - When safety lever releases the escapement lever.

6.3 Timing Element - Consists of safety release ring assembly, safety lever assembly, movement, timing disc assembly and switch assembly.

6.3.1 To operate properly, the timing element must arm, the movement must run to completion of the set time and the switch firing arms close completely. The microswitch on any movement which has armed, shall be open. The microswitch is not considered part of the timing element.

6.4 Environmental tested units - All units subjected to tests in paragraphs 4.2.5 and 4.3.3.23, shall not be used for war reserve unless specifically approved by the responsible engineering agency (SMUFA-J6200).

6.4.1 Process or screening tests - If the timers are run for any reason subsequent to the lubrication of the escapement mechanism per drawing F8595149, this lubrication must be reapplied. Additional lubrication of other areas should also be considered where rework, screening or special testing of the timers is performed.

6.5 Screening of lots - Caution shall be exercised so no damage occurs to components during static load and torque testing.

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6.6 Temperature stabilization of timers may be checked by a thermal measuring device in good thermal contact with the largest centrally located internal mass. A simulated timer may be used for this purpose.

6.7 Data card.-Data card information shall be as specified in MIL-STD-1167.

6.8 Ordering Data.-Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing.

6.9 This specification includes the requirements of Purchase Description FA-PD-M1-2320, Rev. 5 dated 30 November 1963.

Custodian:

Army - MU

Preparing Activity:

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